ELSEVIER

Chemical Engineering Science 61 (2006) 3663–3673

Chemical Engineering Science

www.elsevier.com/locate/ces

Modeling of the evolution with length of bubble size distributions in bubble columns

Sergio Bordel, Rafael Mato, Santiago Villaverde*

Department of Chemical Engineering and Environmental Technology, University of Valladolid, Prado de la Magdalena s/n, 47005, Valladolid, Spain

Received 9 March 2005; received in revised form 14 December 2005; accepted 22 December 2005
Available online 28 February 2006

Abstract

Many of the existing methods, for the determination of the specific interfacial area in bubble columns, consider the column in a dynamic equilibrium between bubble coalescence and breaking-up. The aim of this work is to study if this consideration can be considered true for low superficial gas velocities. Two existing models have been chosen from literature in order to predict the break-up [Wang, T., Wang, J., Jin, Y., 2003. A novel theoretical breakup kernel function for bubbles/droplets in a turbulent flow. Chemical Engineering Science 58, 4629–4637] and the coalescence [Lehr, F., Millies, M., Mewes, D., 2002. Bubble size distributions and flow fields in bubble columns. A.I.Ch.E. Journal 48, 2426] rates. In order to confirm the validity of the models, predictions were compared with experimental results obtained by image analysis. Several simulations were performed for different superficial gas velocities and initial bubble size distributions. The column length needed to reach dynamic equilibrium was calculated for each simulation. The results show that the necessary length to reach the dynamic equilibrium does not depend on the shape of the initial distribution, but essentially on its Sauter mean diameter. The necessary length to reach the dynamic equilibrium is very important for low superficial gas velocities. The assumption that the entire column is in dynamic equilibrium is in general not valid. Therefore, the initial Sauter mean diameter and the total column length are important parameters for the determination of the specific interfacial area.
© 2006 Elsevier Ltd. All rights reserved.

Keywords: Bubble; Bubble columns; Modeling; Simulation; Size distribution; Evolution

1. Introduction

Bubble columns are widely used to carry out mass transfer between a gas and a liquid phase. The mass transfer between the two phases takes place at the interface. Therefore the knowledge of the interfacial area per volume unit is fundamental for the design and operation of bubble columns. The interfacial area per volume unit is a quantity related to the hold up (the gas volume fraction in the column), and to the size distribution of the bubbles.

Two main approaches exist for the determination of interfacial area. The first one consists on the development of empirical correlations for the calculation of the interfacial area and the hold up as a function of certain operational parameters, such as gas surface velocity, and some physical properties of the gas and liquid phases, such as density, viscosity or surface tension.

Several empirical correlations have been proposed by Hughmark (1967), van Dierendronck (1970), Akita and Yoshida (1974), Kumar et al. (1976), Idogawa et al. (1986), Kawase et al. (1987), and Wilkinson (1991). All these correlations provide diverging results that often do not fit to experimental data.

None of these empirical correlations take in account the influence of the size distribution of the bubbles formed at the gas distribution system. It is assumed that the bubble size distribution and the concentration of bubbles in the column are absolutely determined by the parameters included in the correlations, such as the physical properties of the fluids, the superficial velocity of the gas and, in some cases, the column diameter. This assumption is essentially true just at the dynamic equilibrium region, where for any bubble size the rates at which bubbles are formed and broken are equal, i.e., the number of

* Corresponding author. Tel.: +34 983 423 656; fax: +34 983 423 013.
E-mail address: svillave@iq.uva.es (S. Villaverde).

0009-2509/$ - see front matter © 2006 Elsevier Ltd. All rights reserved.
doi:10.1016/j.ces.2005.12.035

bubbles of any size remains constant and bubble size distribution does not change with length.

The second approach for the determination of the interfacial area is more phenomenological and takes in account the coalescence and breaking up of bubbles within the column. Prince and Blanch (1990) were the first to propose and validate a model for calculating bubble coalescence and breakup rates. This is the model that Pohorecki et al. (2001) used for predicting the bubble size distribution corresponding to the dynamic equilibrium region, for several gas–liquid systems. Other models for bubble coalescence and breakup have been proposed more recently, such as those from Chesters (1991), Luo (1993), Luo and Svendsen (1996), Colella et al. (1999), Hagesaether et al. (2002), Lehr et al. (2002), (Wang et al., 2003, 2004). Millies and Mewes (1999) and Lehr and Mewes (2001) proposed several methods to find an approximated analytical solution for bubble size distribution at the dynamic equilibrium region. Wang et al. (2005) and Chen et al. (2005) found that the results predicted by the different coalescence and breakup kernels are quite different. The authors compared several breakup and coalescence models. The best results were obtained with the combination of the breakup kernel of Wang et al. (2003) and the coalescence model of Lehr et al. (2002).

According to Millies and Mewes (1999), there are four regions that appear within any bubble column:

I. The region of primary bubbles formed at the sparger. These bubbles move very fast as a result of the inertia of the gas and therefore break immediately after getting into the column.

II. The region of secondary bubbles, resulting from the breaking of primary bubbles. Secondary bubbles coalesce and break up till reaching a dynamic equilibrium.

III. The dynamic equilibrium region.

IV. The separation region at the top of the column.

The first and fourth regions are too small for influencing the global hold up and interfacial area. However, the influence of the region of secondary bubbles on the global interfacial area might be not negligible as well. The region of secondary bubbles may be of great significance in systems with low superficial gas velocities such as bioreactors used for gas cleaning operating with superficial gas velocities about $0.01\,\mathrm{m\,s^{-1}}$, which are the authors' main subject of study. The presence of electrolytes in the liquid phase reduces the rate of bubble coalescence (Marrucci and Nicodemo (1967)). Therefore, when the liquid phase is an electrolyte solution, the column length to reach the dynamic equilibrium is even higher.

However most of the reported works assume that the dynamic equilibrium region corresponds to the entire column length. The goal of this work is to estimate the impact of the region of secondary bubbles on the interfacial area of a bubble column.

A small number of simulations of the size distribution evolution have been reported in the bibliography, such as those by Shimizu et al. (2000) and Colella et al. (1999). In both works, the bubble sizes are discretized in less than 30 classes. Lehr et al. (2002) and Chen et al. (2005) coupled the bubble population balance with an eulerian simulation of the flow field.

This work is limited to columns working in the homogeneous regime with low superficial gas velocities and therefore without the formation of big bubbles.

The necessary length to reach the dynamic equilibrium is calculated for different superficial gas velocities and initial size distributions. The selected breakup model is the one proposed by Wang et al. (2003). The coalescence rates have been calculated following Lehr et al. (2002).

2. Fundamental equations of bubble columns

A set of bubbles can be characterized by its concentration n, which is the total number of bubbles per volume unit, and its size distribution function, $f(d_b)$, defined as the limit of the fraction of bubbles with diameters between d_b and $d_b + \Delta d$, when Δd tends to zero.

The hold up and the specific interfacial area of a set of bubbles follow Eqs. (1) and (2):

$$H = n\,\frac{\pi}{6}\int_0^\infty d_b^3 f(d_b)\,\mathrm{d}d_b, \tag{1}$$

$$a = n\pi\int_0^\infty d_b^2 f(d_b)\,\mathrm{d}d_b. \tag{2}$$

According to McGinnis and Little (2002), a bubble size distribution is characterized by its Sauter mean diameter (d_s), which is the diameter of a sphere having the same volume-to-surface ratio as the distribution of bubbles. The reason of this choice it is that Sauter diameter is the most representative diameter for mass transfer calculations.

$$d_S = 6\,\frac{H}{a}. \tag{3}$$

The superficial velocity of the gas (U), which is the total gas flow divided by the column section, is related to the size distribution of bubbles and their rising velocities (u_i), the rising velocity depends on bubble diameter (d_b) (Eq. (4)):

$$U = n\,\frac{\pi}{6}\int_0^\infty u(d_b) d_b^3 f(d_b)\,\mathrm{d}d_b. \tag{4}$$

Eqs. (1) and (2) can be rewritten as follows (Eqs. (5) and (6)), if Eq. (4) is used to express the total concentration of bubbles as a function of surface velocity.

$$H = U\,\frac{6\int_0^\infty f(d_b) d_b^3\,\mathrm{d}d_b}{\int_0^\infty f(d_b) u(d_b) d_b^3\,\mathrm{d}d_b}, \tag{5}$$

$$a = U\,\frac{6\int_0^\infty f(d_b) d_b^2\,\mathrm{d}d_b}{\int_0^\infty f(d_b) u(d_b) d_b^3\,\mathrm{d}d_b}. \tag{6}$$

By considering a discrete number of sizes, Eqs. (5) and (6) can be written as follows (Eq. (7) and (8)):

$$H = U\,\frac{\sum_i x_i d_i^3}{\sum_i x_i u_i d_i^3}, \tag{7}$$

$$a = U\,\frac{6\sum_i x_i d_i^2}{\sum_i x_i u_i d_i^3}, \tag{8}$$

where d_i is the diameter of bubbles 'i', and x_i is the number of bubbles 'i' divided by the total number of bubbles.

According to these equations, hold up and specific interfacial area depend on the bubbles size distribution and the velocity of rising bubbles. The velocity of a single rising bubble is a function of its diameter and the physical properties of gas and liquid.

There are different equations in the literature for predicting the rising velocity of bubbles. Prince and Blanch (1990) used the Eq. (9) proposed by Clift (1978), wherein the rising velocity of bubbles is highly dependant on surface tension for small bubbles, and there is no influence of viscosity.

$$u = \left[\frac{2.14\sigma}{\rho_l d_b} + 0.505 g d_b\right]^{1/2}. \tag{9}$$

According to McGinnis and Little (2002) this equation is valid only for bubbles of big size, because small bubbles behave as rigid spheres and their rising velocity do not follow Eq. (9). They proposed two different equations for simulating the rising velocity of bubbles within an air–water system. One equation for bubbles with diameters under 2.6 mm (Eq. (10)):

$$u = \left[\frac{4 d_b g (1 - \rho_g/\rho_l)}{3 C_D}\right]^{1/2}, \tag{10}$$

where C_D is the drag coefficient expressed as

$$C_D = \frac{24}{Re} + \frac{3}{\sqrt{Re}} + 0.34 \tag{11}$$

and Re, the Reynolds number:

$$Re = \frac{u d_b \rho_l}{\mu} \tag{12}$$

and a second equation for bigger bubbles (Eq. (13)). Eqs. (10) and (13) have been selected for calculating the rising velocities of bubbles in this work.

$$u = \left[\frac{2\sigma}{(\rho_l + \rho_g) d_b} + 0.5 g d_b\right]^{1/2}. \tag{13}$$

3. Model development

In order to calculate the evolution with column length of the bubble size distribution, it is necessary to calculate the population balance of bubbles 'i' over a section of the column. For this balance it is assumed that the density of the gas does not change with length, which is a reasonable assumption for small columns. On stationary state, the concentration of bubbles is not time dependant and the concentration of bubbles 'i' will follow the Eq. (14):

$$u_i \frac{\partial n_i}{\partial z} = C_{ai} - C_{di} + B_{ai} - B_{di}, \tag{14}$$

C_{ai} is the rate of apparition of bubbles 'i' as a result of coalescence of smaller bubbles, C_{di} it is the rate of disappearing of bubbles 'i' as a result of their coalescence, B_{ai} it is the rate of apparition because of the breakup of bigger bubbles, and finally B_{di} is the rate of disappearing of bubbles 'i' as a result of their breakup (Tsang et al., 2004).

3.1. Coalescence model

All the existing coalescence models follow Prince and Blanch (1990) considering that the coalescence rate between bubbles 'i' and 'j' is equal to the collision rate (θ_{ij}) multiplied by a collision efficiency coefficient (λ_{ij}).

$$C_{ij} = \theta_{ij} \lambda_{ij}. \tag{15}$$

In this work, two kinds of collisions will be considered, the turbulent collisions and the buoyancy driven collisions, which are due to the different rising velocities.

The turbulent collision rate is given by Eq. (16):

$$\theta_{ij}^{\mathrm{T}} = n_i n_j S_{ij} (u_{ti}^2 + u_{tj}^2)^{0.5}. \tag{16}$$

The buoyancy collision rate is obtained from Eq. (17):

$$\theta_{ij}^{B} = n_i n_j S_{ij} |u_i - u_j|. \tag{17}$$

In Eqs. (16) and (17) S_{ij} is the collision area given by Eq. (18):

$$S_{ij} = \frac{\pi}{4} (d_{bi} + d_{bj})^2. \tag{18}$$

The velocity u_t is the turbulent velocity of bubbles, which is predicted using the Kolmogrov theory (1941) for homogeneous turbulence. According to Prince and Blanch u_t is given by Eq. (19):

$$u_{ti} = 1.4 \varepsilon^{1/3} d_{bi}^{1/3}. \tag{19}$$

The symbol ε is the rate of energy dissipation by unit of liquid mass. If agitation is only due to rising bubbles, ε is given by the following expression (Eq. (20)):

$$\varepsilon = U g. \tag{20}$$

The collision efficiency λ_{ij} is the fraction of collisions producing coalescence between bubbles. Several expressions for λ_{ij} have been proposed in the bibliography. Chen et al. (2005) found that the models proposed by Prince and Blanch (1990), Chesters (1991) and Luo (1993), predict too fast coalescence rates. Lehr et al. (2002) found experimentally that coalescence between bubbles occurs only when the relative velocity of approach perpendicular to the surface of contact is under a certain critical value. For distilled water, this critical value was found to be $0.08\,\mathrm{m\,s^{-1}}$.

The result of Lehr et al. (2002) was used to calculate the collision efficiency. The relative velocity perpendicular to the surface u_p is equal to Eq. (21):

$$u_p = \|\vec{u}_i - \vec{u}_j\| \cos\psi. \tag{21}$$

In Eq. (21), ψ is the angle formed by a vector perpendicular to the surface of collision and the relative velocity of the colliding bubbles. The angle ψ can take values between 0 and $\pi/2$ radians. If we assume that the probability for a collision is equal for all the steradians, the probability density function of ψ is expressed in Eq. (22):

$$f(\psi) = \frac{2\pi sen\psi}{\int_0^{\pi/2} 2\pi sen\psi d\psi} = sen\psi. \tag{22}$$

When the collision is due to buoyancy, the modulus of the relative velocity is equal to the difference between the two rising velocities. Therefore, the condition for coalescence can be derived from Eq. (21) as follows (Eq. (23)):

$$\cos\psi \leqslant \frac{u_{\text{crit}}}{|u_i - u_j|}. \tag{23}$$

If we call ψ_0 to the lower angle for which the inequality becomes true, the fraction of effective collisions will be obtained from Eq. (24):

$$\lambda_{ij}^B = \int_{\psi_0}^{\pi/2} sen\psi \, \mathrm{d}\psi = \cos\psi_0 = \frac{u_{\text{crit}}}{|u_i - u_j|}. \tag{24}$$

If the relative velocity is under the critical velocity then the collision efficiency will be equal to 1.

If the collision is due to the turbulent movement of the bubbles, the relative perpendicular velocity of the bubbles will be given by Eq. (25):

$$\begin{aligned} u_p &= \|\vec{u}_{ti} - \vec{u}_{tj}\| \cos\psi \\ &= \cos\psi \sqrt{\|\vec{u}_{ti}\|^2 + \|\vec{u}_{tj}\|^2 - 2\|\vec{u}_{ti}\|\|\vec{u}_{tj}\| \cos\theta}. \end{aligned} \tag{25}$$

The angle θ is the angle formed by the trajectories of the colliding bubbles. It can take values from 0 to π. It is also assumed that all the relative spatial orientations of the trajectories have the same probability to occur. Therefore the probability density function of θ is expressed in Eq. (26):

$$f(\theta) = \frac{2\pi sen\theta}{\int_0^{\pi} 2\pi sen\theta \, \mathrm{d}\theta} = \frac{sen\theta}{2}. \tag{26}$$

The collisions will result in coalescence whether any of the following conditions is true:

$$\cos\theta \geqslant \frac{\|\vec{u}_{ti}\|^2 + \|\vec{u}_{tj}\|^2 - u_{\text{crit}}^2}{2\|\vec{u}_{ti}\|\|\vec{u}_{tj}\|} \quad \text{or} \quad \cos\psi \leqslant \frac{u_{\text{crit}}}{\|\vec{u}_{ti} - \vec{u}_{tj}\|}. \tag{27}$$

If θ_0 is the bigger angle for which the first inequality becomes true, the fraction of effective collisions is obtained from Eq. (28):

$$\lambda_{ij}^{\mathrm{T}} = \int_0^{\theta_0} \frac{sen\theta}{2} \, \mathrm{d}\theta + \int_{\theta_0}^{\pi} \frac{sen\theta}{2} \int_{\psi_0}^{\pi/2} sen\psi \, \mathrm{d}\psi \, \mathrm{d}\theta. \tag{28}$$

The solution of Eq. (28) is expressed in Eq. (29):

$$\begin{aligned} \lambda_{ij}^{\mathrm{T}} = {} & \frac{1-\zeta}{2} \\ & + u_{\text{crit}} \frac{\|\vec{u}_{ti}\| + \|\vec{u}_{tj}\| \sqrt{\|\vec{u}_{ti}\|^2 + \|\vec{u}_{tj}\|^2 - \|\vec{u}_{ti}\|\|\vec{u}_{tj}\|\zeta}}{2\|\vec{u}_{ti}\|\|\vec{u}_{tj}\|}. \end{aligned} \tag{29}$$

The parameter ζ is equal to Eq. (30):

$$\zeta = \cos\theta_0 = \frac{\|\vec{u}_{ti}\|^2 + \|\vec{u}_{tj}\|^2 - u_{\text{crit}}^2}{2\|\vec{u}_{ti}\|\|\vec{u}_{tj}\|}. \tag{30}$$

The parameter ζ must take values between 1 and -1. Therefore if $\zeta > 1$ its value in Eq. (29) should be 1. The same way if $\zeta < -1$ its value in Eq. (29) should be -1.

The rate of coalescence per volume unit between bubbles 'i' and 'j' can be calculated as follows (Eq. (31)):

$$C_{ij} = \theta_{ij}^B \lambda_{ij}^B + \theta_{ij}^T \lambda_{ij}^{\mathrm{T}}. \tag{31}$$

3.2. Breakup model

The breakup model proposed by Prince and Blanch (1990) is based on the consideration that all bubbles break in two bubbles of the same size. Luo and Svendsen (1996) proposed a model that includes a daughter bubble size distribution. The model of Luo and Svendsen predicts that the daughter bubbles size distribution tends to infinite when the size of one of the daughter bubbles tends to zero, which does not correspond to the reality. Better models have been proposed by Hagesaether et al. (2002) and Wang et al. (2003). In this work it has been chosen the model of Wang.

According to the model of Wang, the breakup rate of bubbles with a size d_i in two bubbles with volumes, one with volume vf_v and the other with volume $v(1-f_v)$ is obtained from Eq. (32):

$$\begin{aligned} b(d_i, f_v) = {} & 0.923(1-H) n_i \varepsilon^{1/3} \\ & \times \int_{\lambda_{\min}}^{d_i} P_b(f_v, d_i, \lambda) \frac{(\lambda + d_i)^2}{\lambda^{11/3}} \, \mathrm{d}\lambda. \end{aligned} \tag{32}$$

In Eq. (32), $P_b(f_v, d_i, \lambda)$ is the probability of breakup of a bubble with a size d_i in two bubbles with volumes, one with volume vf_v and the other with volume $v(1-f_v)$ when it collides with a turbulent eddy of size λ. According to Wang et al. $P_b(f_v, d_i, \lambda)$ is obtained from Eq. (33).

$$P_b(f_v, d_i, \lambda) = \int_0^{\infty} P_b(f_v, d_i, e(\lambda), \lambda) P_e(e(\lambda)) \, \mathrm{d}e(\lambda). \tag{33}$$

The function $P_b(f_v, d_i, e(\lambda), \lambda)$ is the probability of breakup of a bubble with a size d_i in two bubbles, with volumes vf_v and $v(1-f_v)$ when it collides with a turbulent eddy of size λ and energy $e(\lambda)$. $P_e(e(\lambda))$ is the energy distribution of eddies with size λ (Eq. (34)).

$$P_e(e(\lambda)) = \frac{1}{\bar{e}(\lambda)} \exp\left(-\frac{e(\lambda)}{\bar{e}(\lambda)}\right), \tag{34}$$

where the mean kinetic energy of the turbulent eddies is

$$\bar{e}(\lambda) = \frac{\pi}{6}\lambda^3 \rho_l \frac{u_\lambda^2}{2}. \tag{35}$$

The velocity of the turbulent eddies is calculated using Eq. (19).

The probability $P_b(f_v, d_i, e(\lambda), \lambda)$ is calculated as follows:

If $f_v \in (f_{v,\min}, f_{v,\max})$ then $P_b(f_v, d_i, e(\lambda), \lambda) = 1/(f_{v,\max} - f_{v,\min})$ else $P_b(f_v, d_i, e(\lambda), \lambda) = 0$.

The values of $f_{v,\min}$ and $f_{v,\max}$ are calculated from the following Eqs. (36)–(38):

$$f_{v,\min} = \left(\frac{\pi\lambda^3\sigma}{6e(\lambda)d_i}\right)^3, \tag{36}$$

$$c_{f,\max} = \min\left((2^{1/3} - 1), \frac{e(\lambda)}{\pi d_i^2 \sigma}\right), \tag{37}$$

$$c_{f,\max} = (f_{v,\max}^{2/3} + (1 - f_{v,\max})^{2/3} - 1). \tag{38}$$

3.3. Populations balance

The terms in Eq. (14) can be calculated using the previous models in the following way (Eqs. (39)–(42)):

$$C_{ai} = \frac{1}{2}\sum_r \sum_l C_{i,rl}$$
$$\text{where} \begin{cases} C_{i,rl} = C_{rl} & \text{if } v_r + v_l = v_i, \\ C_{i,rl} = 0 & \text{if } v_r + v_l \neq v_i, \end{cases} \tag{39}$$

$$C_{di} = \sum_j C_{ij}, \tag{40}$$

$$B_{di} = \int_0^{0.5} b(f_v, d_i)\,\mathrm{d}f, \tag{41}$$

$$B_{ai} = 2\sum_{j>i} b(f_v, d_i)\Delta v_j / v_i \quad \text{where } f_v = v_j / v_i. \tag{42}$$

4. Numerical method

To integrate the equations system by a numerical method, the diameter of bubbles was discretized into 200 values between 0 and 10^{-2} m at fixed increments of 5×10^{-5} m. This simplification requires to adjust the diameter of bubbles resulting from breaking and coalescence to the nearest discrete value within those 200 values.

In order to keep a constant gas flow through the column, the rate of appearance of bubbles from breaking of bubbles 'i' must be adjusted so that the volume of the new discretized bubbles equals the volume of the broken ones. Likewise, the appearance rate of gas bubbles resulting from coalescence of bubbles 'i' and 'j' must be adjusted so the volume of disappearing bubbles equals the volume of the new formed ones.

The integrals of the breakup model are discretized and solved numerically for each integration step.

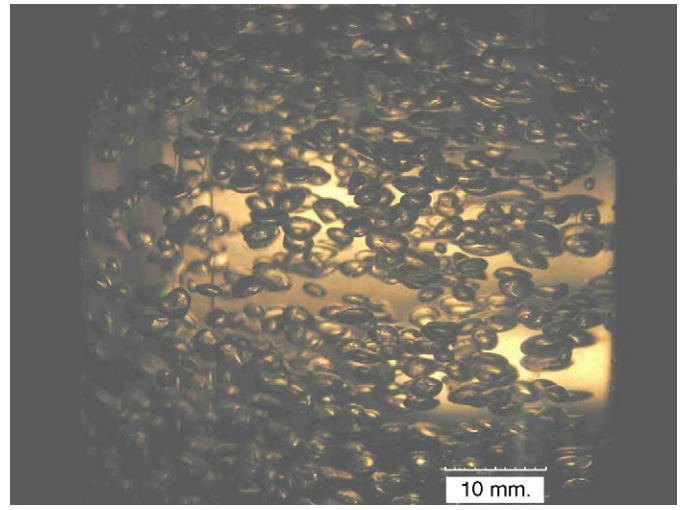


Fig. 1. Bubbles rising in the column at a superficial gas velocity of 0.01 m s^{-1}.

A first order Euler method was selected for solving the system of 200 differential equations resulting from discretization. The integration step was set to 0.5mm to keep the error under 1%.

5. Experimental set up

To test the validity of the mathematical model, experiments were performed in an air-distilled water system, using a column of 10 cm of diameter and 200 cm of length. Bubble size distribution was determined by image analysis, where about 100 bubbles from each photograph were classified in size intervals of 0.5 mm. Each photograph has been used to build an histogram with the frequencies of each bubble size interval.

Photographs could not be taken just over the gas distributor because in this region the gas flow was not homogeneously distributed along the column section. So, the initial distribution has been measured 10 cm above the sparger.

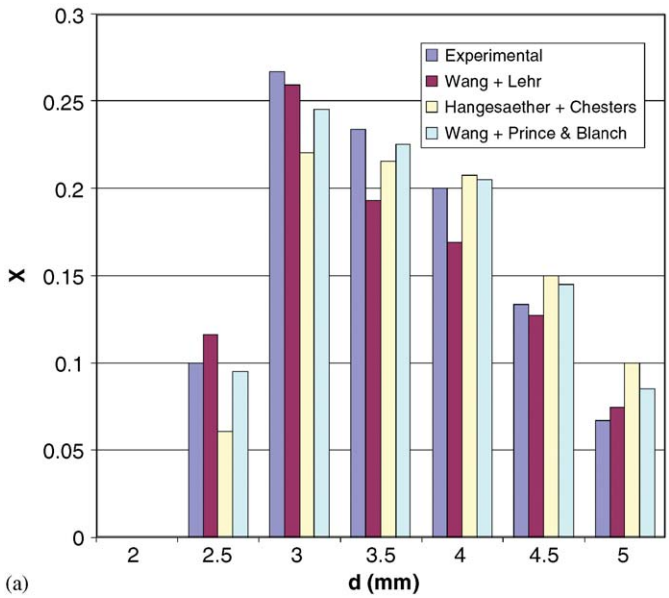
Experiments were performed with two different superficial gas velocities: 0.005 and 0, 01 m s^{-1}. Fig. 1 shows a photograph taken of the bubble column when air was flowing along the column with a superficial velocity of 0.01 m s^{-1}.

6. Results

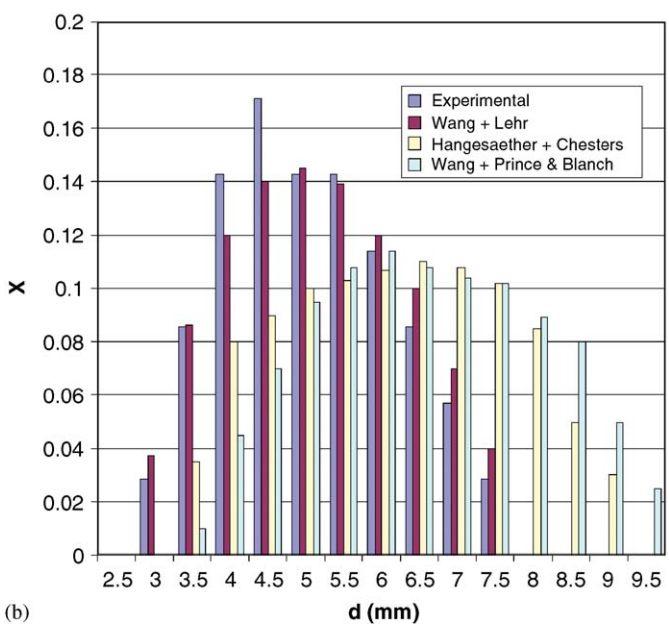
6.1. Comparison between different models

Several combinations of breakup and coalescence models have been tested in order to select the combination that best reproduces the experimental results. The following picture illustrates the results obtained for three different combinations of breakup and coalescence models.

Previous works have shown that the breakup models of Prince and Blanch (1990) and Luo and Svendsen (1996) do not predict correct bubble daughter distributions, therefore they have been discarded. The breakup models of Hagesaether et al. (2002) and



(a)



(b)

Fig. 2. Comparison between the experimental bubble size distribution and the results predicted by three different combinations of breakup and coalescence models: Wang + Lehr, Hagesaether + Chesters and Wang + Prince and Blanch. The simulations correspond to $U = 0.01\,\mathrm{m\,s^{-1}}$; (a) 27 cm, and (b) 145 cm over the initial size distribution.

Wang et al. (2003) were combined with different coalescence models.

The model of Wang has been preferred for the following simulations as it has a lower computational cost.

The graphs above (Fig. 2) show that the coalescence models of Prince and Blanch (1990) and Chesters (1991) predict too fast coalescence rates as it had already been noticed by Chen et al. (2005).

The model of Lehr et al. (2002) proved to give better results and has been chosen for the following simulations.

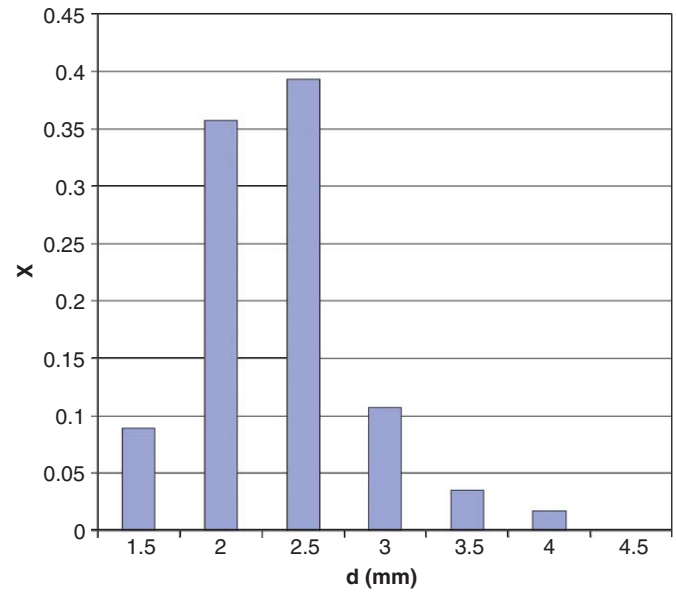


Fig. 3. Initial bubble size distribution for $U = 0.005\,\mathrm{m\,s^{-1}}$. obtained from a photograph taken 10 cm over the column bottom.

6.2. Mathematical model validation

Bubbles were photographed at several column lengths, and the histograms resulting from the measurements were compared with those predicted by the mathematical model.

Bubble size distribution corresponding to the point located 10 cm above the sparger was taken as the initial condition for the integration method. Bubble size distributions at different column lengths were then calculated, and compared to experimental results.

The experimental initial bubble distribution for a superficial velocity of $U = 0.005\,\mathrm{m\,s^{-1}}$ is represented in Fig. 3. The corresponding experimental and calculated bubble size distributions at different column lengths are shown in Fig. 4.

Similar results are presented in Figs. 5 and 6 for a superficial velocity of $U = 0.01\,\mathrm{m\,s^{-1}}$.

A reasonable agreement between experimental and calculated results is observed in both experiments.

6.3. Calculation of the column length necessary to reach dynamic equilibrium

The aim of this work is to make an estimation of the column length necessary to reach the dynamic equilibrium regime. There are two main factors influencing this length: the first one is the initial size distribution, which mainly depends on the gas distributor, and the second factor is the superficial gas velocity.

The influence of the initial size distribution is difficult to evaluate, as there are infinite possible initial distribution functions. Several simulations were performed using a constant superficial gas velocity of $0.015\,\mathrm{m\,s^{1}}$ for a set of different initial size distributions. A total of 14 Dirac functions, which consider that all the bubbles have the same initial size, and 15 Weibull

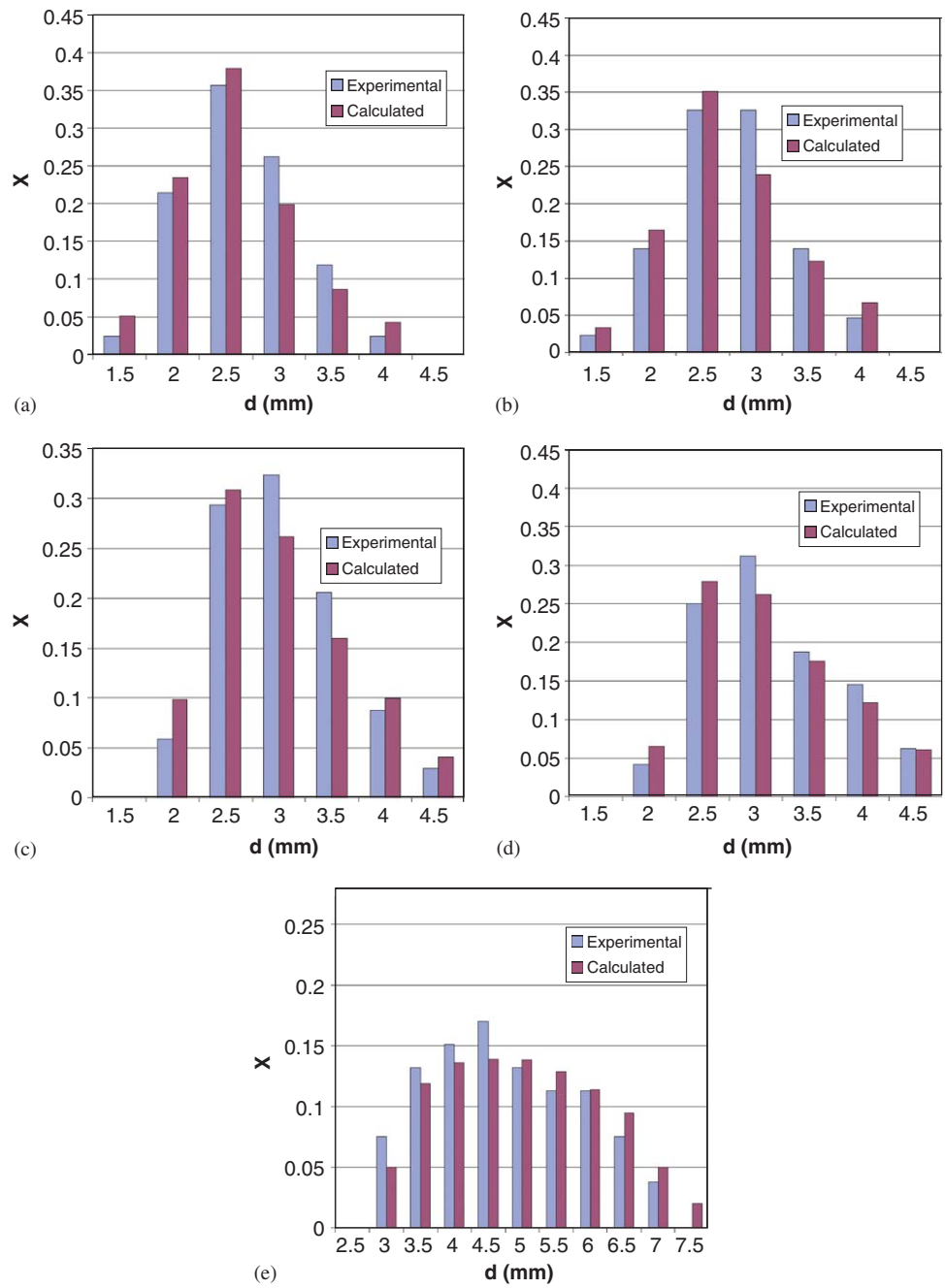


Fig. 4. Bubble size distribution at different column lengths above initial distribution point for $U = 0.005\,\mathrm{m\,s^{-1}}$: (a) 9 cm; (b) 16 cm; (c) 26 cm; (d) 34 cm; and (e) 145 cm.

distributions (Eq. (43)), were used.

$$f(d_b) = r^3 \left(\frac{\pi}{6}\right) d_b^{3c-1} \mathrm{e}^{-\frac{r}{c}\left(\frac{\pi}{6}d_b^3\right)^c}. \tag{43}$$

These functions were chosen because they have similar shapes to the experimental distributions, and because they are easier to manipulate than the more usual normal and log-normal distributions.

As previously explained, bubble size distributions are frequently characterized by their Sauter mean diameter (McGinnis and Little, 2002), given by Eq. (3). In this work, the required criteria to decide if the dynamic equilibrium regime has been reached was a Sauter mean diameter discrepancy of less than 5% respect its asymptotic value, corresponding to an infinite high column.

The parameters and Sauter mean diameters of the 28 initial size distributions used in the simulations are shown in Table 1.

Fig. 7 represents the evolution of the Sauter mean diameter with length for 7 of the 28 performed simulations.

Fig. 8 represents the column length necessary to reach dynamic equilibrium as a function of the deviation of the initial size distribution from the one corresponding to dynamic

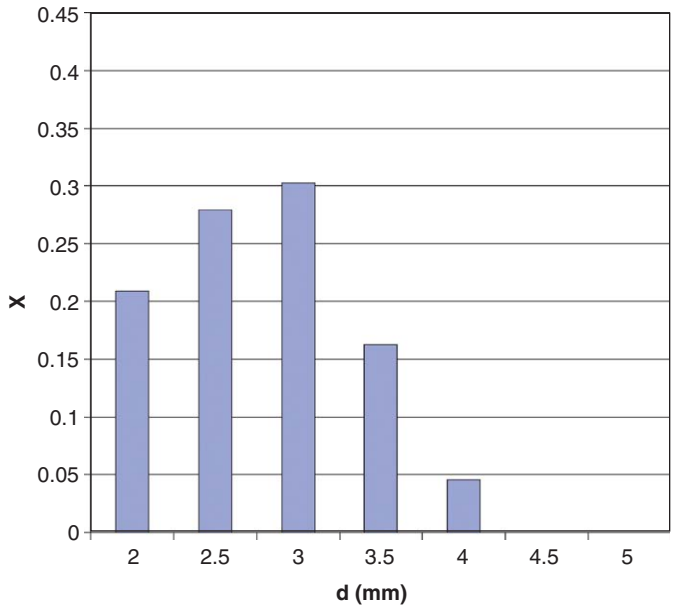


Fig. 5. Initial bubble size distribution for $U = 0.01\,\mathrm{m\,s^{-1}}$ obtained from a photograph taken 10 cm over the column bottom.

Table 1
Parameters and Sauter mean diameters of the 28 initial bubble size distributions used in the simulations

Weibull distributions				Dirac distributions	
No.	r	c	d_s (mm)	No.	d_s (mm)
1	$5 \times 10^{13}\,\mathrm{m}^{-6}$	2.0	7.21	1	3.30
2	$5 \times 10^{15}\,\mathrm{m}^{-6}$	2.0	3.37	2	2.80
3	$2 \times 10^{16}\,\mathrm{m}^{-6}$	2.0	2.68	3	2.00
4	$10^{17}\,\mathrm{m}^{-9}$	2.0	2.05	4	3.60
5	$4 \times 10^{17}\,\mathrm{m}^{-6}$	2.0	1.63	5	4.50
6	$10^{25}\,\mathrm{m}^{-6}$	3.0	2.31	6	1.30
7	$10^{23}\,\mathrm{m}^{-7.5}$	2.5	2.31	7	1.00
8	$5 \times 10^{19}\,\mathrm{m}^{-7.5}$	2.5	3.28	8	7.80
9	$10^{21}\,\mathrm{m}^{-6}$	2.0	0.44	9	8.50
10	$10^{15}\,\mathrm{m}^{-6}$	2.0	4.39	10	4.00
11	$5 \times 10^{25}\,\mathrm{m}^{-10.5}$	3.5	4.90	11	5.30
12	$9 \times 10^{24}\,\mathrm{m}^{-10.5}$	3.5	5.76	12	6.85
13	$1 \times 10^{20}\,\mathrm{m}^{-9}$	3.0	8.22	13	7.20
14	$10^{22}\,\mathrm{m}^{-9}$	3.0	4.95	14	9.50
15	$10^{14}\,\mathrm{m}^{-6}$	2.0	6.44		

equilibrium. This initial deviation (ID) will be represented by the difference between the initial and equilibrium Sauter diameters divided by the equilibrium Sauter diameter (Eq. (37)).

$$ID = \frac{d_{Se} - d_{S0}}{d_{Se}}. \tag{44}$$

According to this definition, Sauter diameters smaller than the equilibrium one (the most usual scenario) mean positive initial deviations and Sauter diameters bigger than the equilibrium one mean negative initial deviations.

A similar pattern is observed in all simulations for both families of initial distributions, showing a greater dependence of the dynamic equilibrium column length with the Sauter diameter of the initial distribution than with its shape.

Fig. 9 shows the dynamic equilibrium column length as a function of the initial Sauter diameter deviation for three different superficial gas velocities.

Figs. 8 and 9, show how for positive initial deviations, when the initial deviation is under 0.5, a linear relationship between Z_e and the initial deviation may be assumed. For initial deviations over 0.5 the column length needed to reach dynamic equilibrium is roughly independent of the initial deviation and depends only on superficial gas velocity.

For negative initial deviations Z_e increases with the deviation from equilibrium, until an asymptotic value. The column length needed to reach the dynamic equilibrium is smaller than for positive initial deviations, which means that the breakup of the bubbles is a faster process than the coalescence.

In order to test the influence of velocity on the dynamic equilibrium column length, a Dirac distribution with $d = 1.3$ mm. was chosen as initial size distribution. This distribution corresponds to an initial deviation over 0.5, which guarantees that the only influence over the equilibrium length is the superficial gas velocity. With this initial size distribution, simulations for six gas superficial velocities (0.005, 0.0075, 0.01, 0.015, 0.02 and $0.03\,\mathrm{m\,s^{-1}}$) were performed. The results of three of the simulations are shown in Fig. 10.

Fig. 11 shows the necessary column length for reaching the dynamic equilibrium as a function of the superficial gas velocity. All the simulations have been performed taking as initial distribution a Dirac function with a Sauter mean diameter of 1.3 mm.

According to these results, the necessary length to reach the dynamic equilibrium is very important for small superficial gas velocities.

7. Conclusions

The chosen models are capable to predict with a reasonable accuracy the evolution of the size distribution in a bubble column operating with low superficial gas velocities. The necessary column length to reach the dynamic equilibrium does not depend on the shape of the initial distribution, but essentially on its Sauter mean diameter. This influence on the initial Sauter diameter is not important when it differs in more than 50% from the Sauter diameter at the dynamic equilibrium. In this case, the equilibrium length only depends on the superficial gas velocity.

The assumption that the entire column is in dynamic equilibrium is in general not valid. Therefore, the initial Sauter mean diameter and the total column length are important parameters for the determination of the specific interfacial area. Particularly, for low superficial gas velocities, the equilibrium length is extremely important.

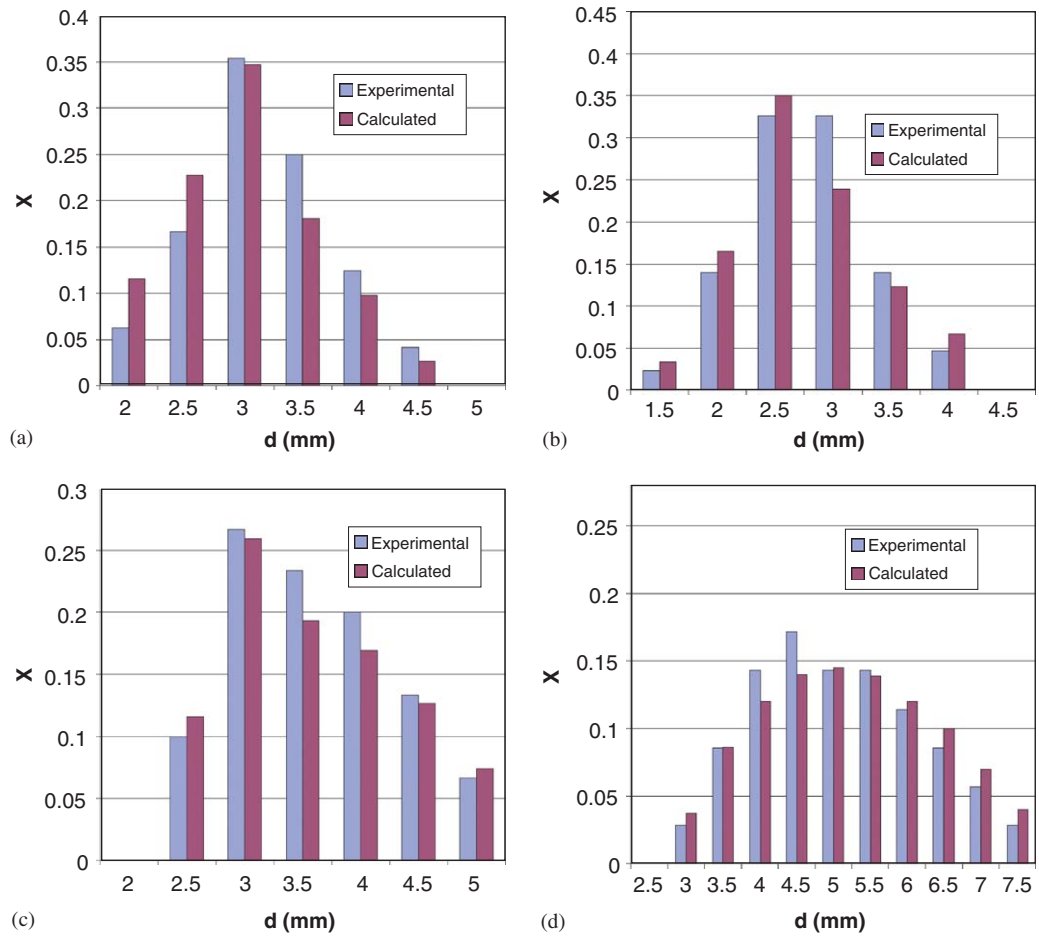


Fig. 6. Bubble size distribution at different column lengths above initial distribution point for $U = 0.01\,\mathrm{m\,s^{-1}}$: (a) 6 cm; (b) 16 cm; (c) 27 cm; and (d) 145 cm.

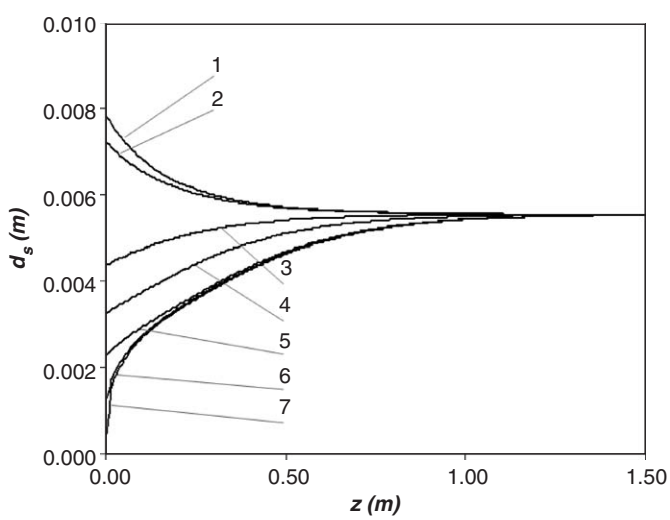


Fig. 7. Evolution of the Sauter mean diameter along the column for a superficial gas velocity of $0.015\,\mathrm{m\,s^{-1}}$, for different initial bubble size distributions: (1) Dirac 8; (2) Dirac 13; (3) Weibull 10; (4) Weibull 8; (5) Weibull 6; (6) Dirac 6; and (7) Weibull 9.

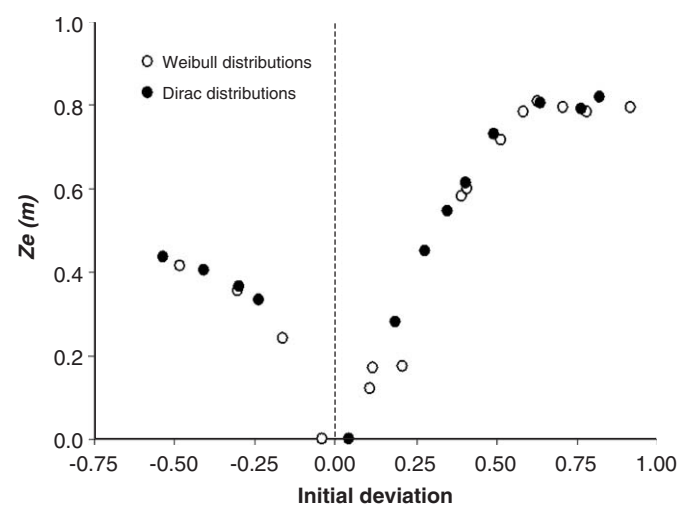


Fig. 8. Column length necessary to reach the dynamic equilibrium as a function of the initial Sauter diameter deviation, for a superficial gas velocity of $0.015\,\mathrm{m\,s^{-1}}$.

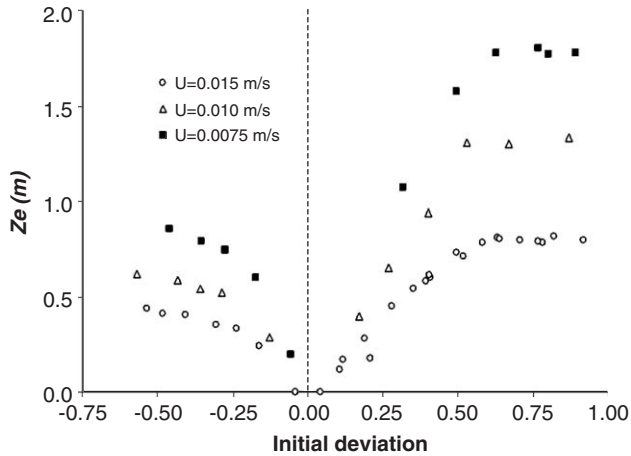


Fig. 9. Dynamic equilibrium length as a function of the initial Sauter diameter deviation for three different superficial gas velocities.

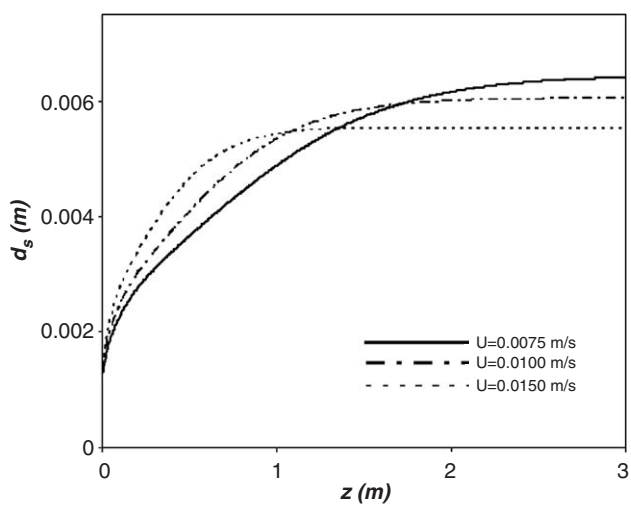


Fig. 10. Evolution of Sauter mean diameter along the column for different superficial gas velocities for an initial Dirac distribution with $d = 1.3$ mm.

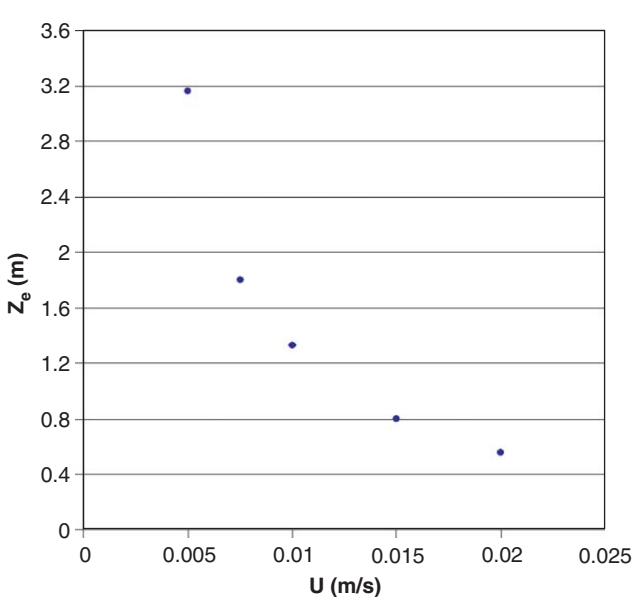


Fig. 11. Column length necessary to reach the dynamic equilibrium as a function of the superficial gas velocity for an initial Dirac distribution with $d = 1.3$ mm.

Notation

a	gas–liquid interfacial area per unit volume, m^{-1}
B_{ie}	number of bubbles of class 'i' that break up per time and volume units as a result of collisions with eddies 'e', $s^{-1}\,m^{-3}$
B_{ai}	rate of apparition of bubbles of class 'i' per volume unit as a result of the break up of bigger bubbles, $s^{-1}\,m^{-3}$
B_{di}	disappearing rate of bubbles of class 'i' per volume unit as a result of the break up, $s^{-1}\,m^{-3}$
c	parameter of the Weibull distribution
C_{ij}	number of coalescence events per unit of time and volume between bubbles of classes 'i' and 'j', $s^{-1}\,m^{-3}$
C_{ai}	rate of apparition per volume unit of bubbles of the class 'i' as a result of the coalescence of smaller bubbles, $s^{-1}\,m^{-3}$
C_{di}	disappearing rate per volume unit of bubbles of class 'i' as a result of coalescence, $s^{-1}\,m^{-3}$
d_b	bubble diameter, m
d_i	diameter of the bubbles of the size class 'i', m
d_s	Sauter mean diameter, m
$e(\lambda)$	eddy energy, J
$f(d)$	size distribution function, m^{-1}
f_v	fraction of the parent bubble corresponding to the smaller daughter bubble
g	gravity acceleration, $m\,s^{-2}$
h_0	initial size of the liquid film between two bubbles, m
h_f	size of the liquid film between twobubbles when coalescence occurs, m
H	gas hold up
ID	initial deviation
k	wave number of the eddies, m^{-1}
n	total number of bubbles per volume unit, m^{-3}
n_i	number of bubbles of the size class 'i' per volume unit, m^{-3}
n_e	number of eddies per volume unit, m^{-3}
P_b	Probability density for the bubble break up
r	parameter of the Weibull distribution
Re	Reynolds' number
S_{ij}	collision area of bubbles of classes 'i' and 'j', m^2
S_{ie}	collision area of bubbles of class 'i' and eddies, m^2
t_{ij}	time required for bubble coalescence, s
u	rising velocity of bubbles, $m\,s^{-1}$
u_i	rising velocity of bubbles of the size class 'i', $m\,s^{-1}$
u_{crit}	critical velocity for coalescence, $m\,s^{-1}$
u_t	turbulent velocity, $m\,s^{-1}$
u_λ	velocity of a turbulent eddy, ms^{-1}
U	gas superficial velocity, $m\,s^{-1}$
v	volume of the bubbles, m^3
x_i	fraction of bubbles 'i',
z	length, m
Z	length of the region of secondary bubbles, m

Greek letters

ε power dissipated per unit mass of liquid, $W\,kg^{-1}$
θ_{ij} collisions frequency between bubbles of classes 'i' and 'j', $s^{-1}\,m^{-3}$
θ Angle formed between the surface of collision and the relative velocity of two colliding bubbles.
κ_{ie} collisions efficiency between bubbles and eddies
λ eddy diameter, m
λ_{ij} coalescence collision efficiency between bubbles of classes 'i' and 'j'
μ viscosity of the liquid, $kg\,m^{-1}\,s^{-1}$
ρ_l density of the liquid, $kg\,m^{-3}$
ρ_g density of the gas, $kg\,m^{-3}$
σ surface tension, $N\,m^{-1}$
ψ angle formed by the trajectories of two colliding bubbles

Acknowledgements

The authors wish to thank the financial support of the Ministry of Education and Science of Spain (contract PPQ2003-09044) and the European Commission (contract HPMD CT 2001-00106).

References

Akita, K., Yoshida, F., 1974. Bubble size, interfacial area, and liquid phase mass transfer coefficient in bubble columns. Industrial and Engineering Chemistry, Process Design and Development 13 (1), 89–91.

Chen, P., Sanyal, J., Dudukovic, M.P., 2005. Numerical simulation of bubble column flows: effect of different breakup and coalescence closures. Chemical Engineering Science 60, 1085–1101.

Chesters, A.K., 1991. The modeling of Coalescence processes in fluid–liquid dispersions. Transactions of the Institution of Chemical Engineers 69, 259–270.

Clift, R., Grace, J.R., Weber, M.E., 1978. Bubbles Drops and Particles. Academic Press, New York.

Colella, D., Vinci, D., Bagatin, R., Masi, M., 1999. A study on coalescence and breakage mechanisms in three different bubble columns. Chemical Engineering Science 54, 4767–4777.

van Dierendronck, L.L., 1970. Vergrotingsregels voor gasbelwassers. Ph.D. Thesis, University of Twente, Esnschede.

Hagesaether, L., Jakobsen, H.A., Svendsen, H.F., 2002. A model for turbulent binary breakup of dispersed fluid particles. Chemical Engineering Science 57, 3251–3257.

Hughmark, G.A., 1967. Holdup and mass transfer in bubble columns. Industrial and Engineering Chemistry. Process design and development 6 (2), 218–220.

Idogawa, K., Ikeda, K., Fukuda, T., Morooka, S., 1986. Behaviour of bubbles of the air–water system in a column under high pressure. International Chemical Engineering 26 (3), 468–474.

Kawase, Y., Halard, B., Moo-Young, M., 1987. Theoretical prediction of volumetric mass transfer coefficients in bubble columns for newtonian and non newtonian fluids. Chemical Engineering Science 42, 1609–1617.

Kolmogorov, A.N., 1941. The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers. Doklady Akdemii Nauk SSSR 30 (6), 301–305.

Kumar, A., Degaleesan, T., Laddha, G.S., Hoelscher, H.S., 1976. Bubble swarm characteristics in bubble columns. Canadian Journal of Chemical Engineering 54 (6), 503–508.

Lehr, F., Mewes, D., 2001. A transport equation for the interfacial area density applied to bubble columns. Chemical Engineering Science 56, 1159–1166.

Lehr, F., Millies, M., Mewes, D., 2002. Bubble size distributions and flow fields in bubble columns. A.I.Ch.E. Journal 48, 2426.

Luo, H., 1993. Coalescence, breakup and liquid circulation in bubble column reactors. D.Sc. Thesis, Norwegian Institute of Technology.

Luo, H., Svendsen, H.F., 1996. Theoretical model for drop and bubble breakup in turbulent dispersions. A.I.Ch.E. Journal 42, 1225–1233.

Marrucci, G., Nicodemo, L., 1967. Coalescence of gas bubbles in aqueous solutions of inorganic electrolytes. Chemical Engineering Science 22, 1257–1265.

McGinnis, D.F., Little, J.C., 2002. Predicting diffused-bubble oxygen transfer rate using the discrete-bubble model. Water Research 36, 4627–4635.

Millies, M., Mewes, D., 1999. Interfacial area density in bubbly flow. Chemical Engineering and Processing 38, 307–319.

Pohorecki, R., Moniuk, W., Bielski, P., Zdrójkoski, A., 2001. Modelling of the coalescence redispersion processes in bubble columns. Chemical Engineering Science 56, 6157–6164.

Prince, M.J., Blanch, H.W.B., 1990. Coalescence and break-up in air-sparged bubble columns. A.I.Ch.E. Journal 36 (10), 1485–1499.

Shimizu, K., Takada, S., Minekawa, K., Kawase, Y., 2000. Phenomenological model for bubble column reactors: prediction of gas hold-ups and volumetric mass transfer coefficients. Chemical Engineering Journal 78, 21–28.

Tsang, Y.H., Koch, Y., Koch, D.L., 2004. Bubble size dependence of the critical electrolyte concentration for inhibition of coalescence. Journal of Colloid and Interface Science 275, 290–297.

Wang, T., Wang, J., Jin, Y., 2003. A novel theoretical breakup kernel function for bubbles/droplets in a turbulent flow. Chemical Engineering Science 58, 4629–4637.

Wang, T., Wang, J., Jin, Y., 2004. An efficient numerical algorithm for A novel theoretical breakup kernel function of bubble/droplet in a turbulent flow. Chemical Engineering Science 59, 2593–2595.

Wang, T., Wang, J., Jin, Y., 2005. Population balance model for gas–liquid flows: influence of bubble coalescence and breakup models. Industrial Engineering Chemistry Research 44, 7540–7549.

Wilkinson, P.M., 1991, Physical aspects and scale-up of high-pressure bubble columns. Ph.D. Thesis, University of Groningen.